

From Conflicting Objectives to Sequential Success: Mitigating Belief Bias in LLMs with S-CoT Distillation and Post-Training Unlearning

Abstract

Large Language Models (LLMs) exhibit significant belief bias in syllogistic reasoning, prioritizing plausibility over logical validity. We first show that a dual-objective approach, combining Symbolic Chain-of-Thought (S-CoT) distillation with activation steering, fails catastrophically. This simultaneous training creates an objective conflict that severely degrades reasoning accuracy. Based on this finding, we propose a novel, two-stage "Train-then-Debias" paradigm. We first distill S-CoT into a high-capacity model (Qwen-8B) to establish a strong, accurate baseline. We then apply a post-training intervention, Selective Disentanglement Unlearning (SDU), which successfully neutralizes the residual belief bias with a minimal and acceptable accuracy trade-off. Our work provides empirical evidence for the superiority of this sequential approach, establishing a viable path toward models that are both logically robust and unbiased.¹

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities, yet they often inherit human-like cognitive biases (Dasgupta et al., 2022). A prominent example is **belief bias**: the tendency to judge the validity of an argument based on the believability of its conclusion rather than its logical structure (Evans et al., 1983). For instance, a model might incorrectly accept an invalid argument if its conclusion is plausible, or reject a valid argument if its conclusion is implausible. This flaw undermines the reliability of LLMs in high stakes domains requiring formal, content independent reasoning, such as law, science, and medicine.

The SemEval-2025 Task 11 dataset provides a structured environment to measure this bias using syllogisms, quantifying it via the **Content Effect**

(CE) score—the accuracy difference between plausible and implausible items. A high CE indicates that the model relies on plausibility heuristics rather than logical form.

To tackle this, we investigate two promising techniques:

1. **Symbolic Chain-of-Thought (S-CoT)**: We move beyond simple classification by prompting models to generate explicit, symbolic reasoning traces (e.g., "All A are B..."). This aims to force the model to engage with the logical structure, drawing inspiration from works on CoT (Wei et al., 2022) and symbolic reasoning (Nye et al., 2021). We employ knowledge distillation (Hinton et al., 2015; Hsieh et al., 2023) to teach this skill to smaller student models.
2. **Representation Engineering**: We aim to directly identify and modify the internal model representations associated with bias. We explore two variants: (a) **Activation Steering** (Turner et al., 2023; Zou et al., 2023), applied as a penalty loss during training, and (b) **Selective Disentanglement Unlearning (SDU)** (Liu et al., 2024), a surgical post training intervention.

Our research follows a "failure and redemption" narrative. We first hypothesized that we could mitigate bias by combining S-CoT distillation and activation steering in a simultaneous, dual-objective training process on Gemma models (2B, 4B). As detailed in Section 4.1, this approach failed catastrophically. While activation steering was highly effective at its narrow task (86% bias reduction), it created a fundamental conflict with the reasoning objective, causing accuracy to plummet by 20% (from 69.86% to 49.32%).

This failure revealed two critical insights:

¹Code and data available at: <https://github.com/iDheer/SemEval11>

1. **Dual-Objective Conflict:** The optimization goals of learning complex S-CoT reasoning and avoiding bias-related representations are contradictory when applied simultaneously. The steering loss dominates, preventing the model from learning to reason.
2. **Model Capacity Threshold:** S-CoT generation is a far more complex task than simple classification. Our experiments show that models with 2B-4B parameters are insufficient, failing to outperform even a simple RoBERTa baseline.

Based on this failure analysis, we formulated a new hypothesis: reasoning capability and bias mitigation must be decoupled and addressed sequentially. This led to our three stage methodology, depicted in Figure 1:

- **Stage 0 (Capability Acquisition - Baseline Training):** We first validate our S-CoT distillation approach on a high-capacity model (Qwen-8B). This model proves large enough to master the symbolic reasoning task, achieving a strong 81.5% accuracy baseline, significantly outperforming the RoBERTa baseline.
- **Stage 1 (Bias Refinement):** We begin with a capable, pre-trained model that exhibits undesirable belief bias despite its high performance. The goal of this stage is to isolate and prepare the model for a targeted debiasing procedure.
- **Stage 2 (SDU Intervention):** We apply a post-training intervention, Selective Disentanglement Unlearning (SDU) (Liu et al., 2024), to surgically identify and remove the model’s residual belief bias.

This sequential paradigm was a clear success. The SDU intervention successfully neutralized the model’s bias (shifting CE from -0.0314 to +0.0098) while incurring only a minimal, acceptable 1.4% drop in overall accuracy.

Our key contributions are:

1. Empirical evidence of the dual objective conflict that arises from simultaneously training for reasoning and applying activation steering.
2. Demonstration of a model capacity threshold (approx. 8B parameters) for S-CoT distillation to be effective in this symbolic reasoning task.

3. The first (to our knowledge) adaptation and application of Selective Disentanglement Unlearning (SDU) to mitigate syllogistic belief bias.
4. A successful and validated two stage "Train-then-Debias" paradigm that achieves both high reasoning accuracy and near-perfect bias neutrality.

2 Background and Related Work

2.1 Belief Bias in Humans and LLMs

Belief bias is a long-standing concept in human cognitive psychology (Evans et al., 1983). It describes the phenomenon where individuals are more likely to accept a conclusion as logically valid if it aligns with their pre-existing beliefs, and reject it if it does not, regardless of the logical structure of the premises. This leads to a distinct performance pattern:

- **High Accuracy** on *Valid-Plausible (VP)* and *Invalid-Implausible (II)* items, where logic and belief align.
- **Low Accuracy** on *Valid-Implausible (VI)* and *Invalid-Plausible (IP)* items, where logic and belief conflict.

Recent work has shown that LLMs exhibit strikingly similar human-like content effects, demonstrating a robust belief bias that scales with model size (Dasgupta et al., 2022). Our work uses the SemEval-2025-Task 11 dataset, which is specifically designed to elicit and measure this bias through these four categories.

2.2 Chain-of-Thought and Rationale Distillation

Standard LLM prompting often elicits "System 1" thinking—fast, intuitive, and heuristic-based. Chain-of-Thought (CoT) prompting (Wei et al., 2022) attempts to elicit "System 2" thinking by instructing the model to generate intermediate reasoning steps, leading to improved performance on complex tasks. Nye et al. (2021) further showed that forcing models to generate symbolic traces (like formal logic) enhances interpretability and accuracy in logical tasks.

Our Symbolic-CoT (S-CoT) approach is a minimalist 3-step variant: (1) Abstract concepts to variables, (2) Formulate in first-order logic, (3) Conclude VALID/INVALID. Instead of prompting

at inference time, we use knowledge distillation (Hinton et al., 2015) to fine-tune a smaller student model to generate these S-CoT rationales, inspired by recent work on rationale distillation (Hsieh et al., 2023). The hypothesis is that learning to generate the process of reasoning will build a more robust and less biased model than one simply trained on (input, label) pairs.

2.3 Representation Engineering for Debiasing

This field aims to control model behavior by directly identifying, interpreting, and manipulating a model’s internal activations (Zou et al., 2023).

Activation Steering Turner et al. (2023) demonstrated that model behavior can be "steered" by adding a "bias vector" to activations at inference time. An alternative, which we explore, is to apply this during training as a steering loss. A "bias direction" is extracted from contrastive examples (e.g., $v_{\text{bias}} = \text{mean}(z_{\text{IP}}) - \text{mean}(z_{\text{II}})$), and a loss term $L_{\text{steer}} = \lambda \cdot (1 - \text{cosine_similarity}(z_{\text{current}}, v_{\text{bias}}))$ penalizes the model for activating in this direction. This was the core of our initial, failed methodology.

Machine Unlearning While steering applies a continuous penalty during learning, unlearning techniques aim to surgically remove specific knowledge or biases after training is complete. Liu et al. (2024) proposed "Disentangling Biased Knowledge from Reasoning... via Machine Unlearning," which we adapt as Selective Disentanglement Unlearning (SDU). SDU works by: (1) training a "shadow model" on bias-correlated data to identify bias-sensitive parameters (saliency mask), (2) computing the bias direction in activation space, and (3) applying a one-time, targeted weight update to the original model to "unlearn" the bias while preserving reasoning. This post-training, sequential approach directly addresses the dual-objective conflict we identified in our initial experiments.

3 Dataset and Task Definition

3.1 SemEval-2025-Task 11

We use the dataset from SemEval-2025-Task 11, consisting of 960 syllogisms. Each syllogism comprises two premises and a conclusion. Each item is labeled for both **Validity** (True/False) and **Plausibility** (True/False).

3.2 The Four Categories

The dataset is structured into four categories, representing the conflict (or alignment) between logic and belief. This structure, shown in Table 1, is essential for diagnosing belief bias.

Category	Validity	Plausibility	Example
VP	Valid	Plausible	"All cats are mammals..."
VI	Valid	Implausible	"All flowers are bears..."
IP	Invalid	Plausible	"All doctors are professionals..."
II	Invalid	Implausible	"Some rocks are alive..."

Table 1: The four categories of syllogisms used to measure belief bias. The conflict items (VI and IP) are the most difficult for biased models.

3.3 Evaluation Metrics

Our evaluation relies on two primary metrics:

1. **Accuracy:** The overall percentage of correct validity classifications on the test set. This measures the model’s general reasoning capability.
2. **Content Effect (CE):** The primary metric for belief bias. A score of 0 indicates perfect neutrality.

The Content Effect is calculated as:

$$\text{CE} = \text{Acc}(\text{Plausible}) - \text{Acc}(\text{Implausible})$$

This precise calculation of CE helps to robustly quantify the influence of plausibility on the model’s judgments.

3.4 Data Split

We created a custom balanced 70%/15%/15% stratified split, ensuring all four categories were represented proportionally in the training (671 examples), validation (143 examples), and test (146 examples) sets. The validation set was also used to extract contrastive activation sets (e.g., IP vs. II for initial steering, VP vs. VI for SDU) for bias vector computation. This stratified split is crucial to ensure that the evaluation metrics are not skewed by an imbalance in any specific category.

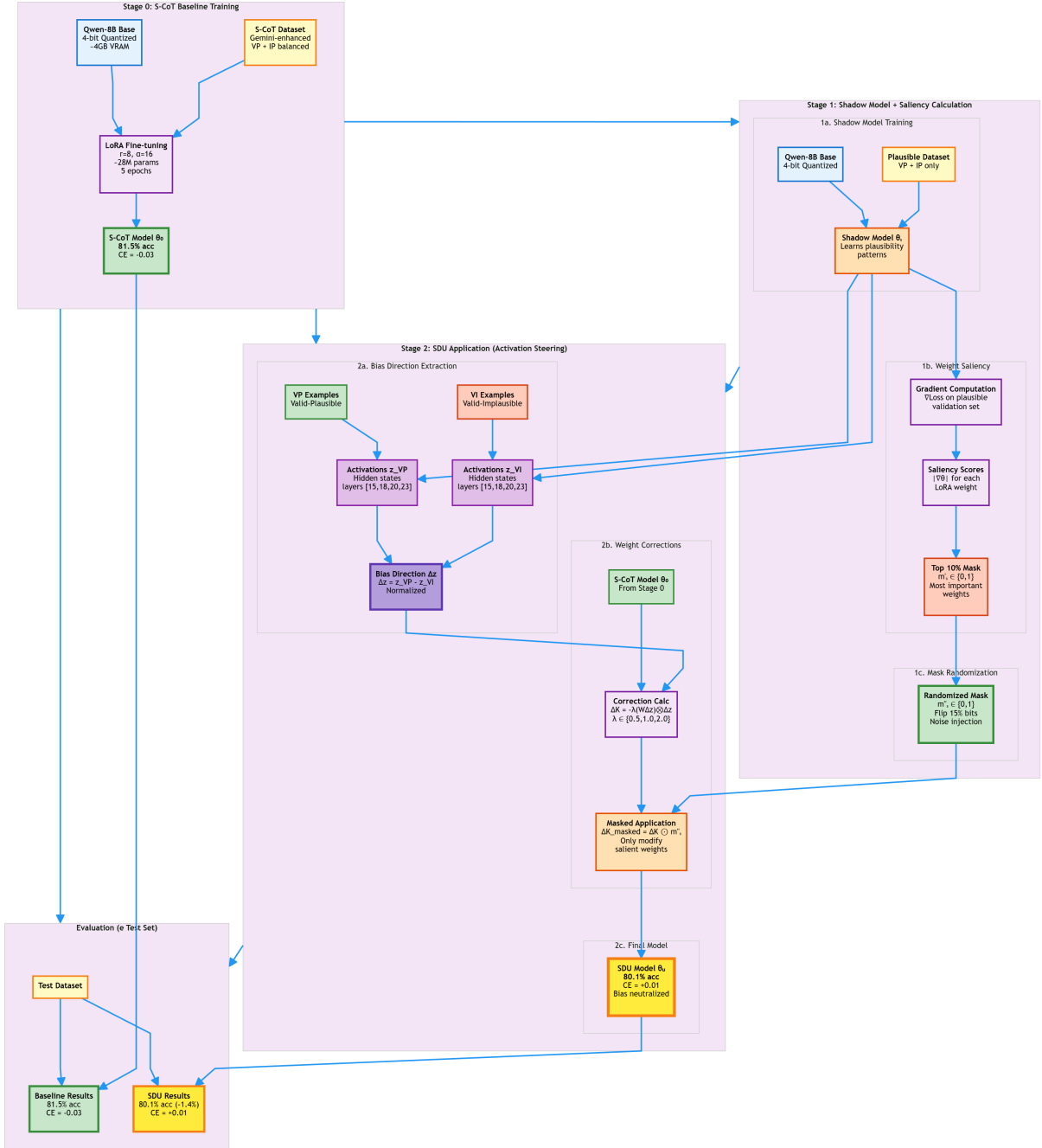


Figure 1: Overview of the Proposed Sequential Architecture: Our methodology involves two main stages. **Stage 0** establishes a capable S-CoT baseline model (Qwen-8B) with high reasoning accuracy. **Stage 1** focuses on *Shadow Model Training and Saliency Calculation* (Blocks 1a, 1b, 1c) to identify plausibility-sensitive parameters. **Stage 2** outlines the *SDU Application* (Blocks 2a, 2b, 2c), which computes the bias direction and applies masked weight corrections to the Stage 0 baseline model, creating a debiased model. Both the baseline and the SDU-corrected models are rigorously evaluated on the Test Dataset (Block 3).

4 Methodology: From Failure to Sequential Success

Our research pivoted dramatically after our initial catastrophic failures. This section details both the failed "simultaneous optimization" approach and our successful "sequential intervention" paradigm, explaining the architectural flow as depicted in Figure 1.

4.1 Part 1: The Initial Hypothesis (Simultaneous Optimization) - The Catastrophic Failure

Our first approach was based on the hypothesis that we could train a model to be both a good reasoner and unbiased at the same time.

S-CoT Teacher and Baseline We utilized Google’s Gemini 2.5 Flash API as our teacher model for generating Symbolic Chain-of-Thought (S-CoT) rationales. The prompt engineering produced a minimalist 3-step S-CoT: (1) Symbolic Abstraction, (2) Logical Formulation, and (3) Final Answer (VALID/INVALID), as exemplified in Appendix A. This format was designed to compel the model towards structured, symbolic reasoning, minimizing extraneous natural language noise.

The teacher achieved 83.75% accuracy (531 out of 634 parsable training examples). Following insights from Hsieh et al. (2023) regarding the detrimental effect of training on erroneous rationales, we rigorously filtered our S-CoT training set to include **only the 531 high-quality, correct examples**.

A RoBERTa-base model, fine-tuned for simple binary classification on the original 671 training examples, served as our initial implicit pattern-matching baseline. It achieved an accuracy of **69.86%** and a **Content Effect of 0.101**.

Experimental Setup (Gemma Models) We conducted five systematic experiments with Gemma-2-2b and Gemma-3-4b-it models.

- **Models:** Gemma-2-2b and Gemma-3-4b-it, fine-tuned with LoRA (rank $r = 16$, alpha $\alpha = 32$).
- **Training Data:** The filtered 531 correct S-CoT traces.
- **Activation Steering Vector Extraction:** The plausibility bias direction (v_{bias}) was extracted from the validation set. This involved contrasting the mean activations of Invalid-Plausible

(IP) examples and Invalid-Implausible (II) examples at specific transformer layers (Layer 20 or 23). The choice of IP vs. II is critical to isolate the bias, as both are logically invalid but differ in plausibility. The vector was normalized.

$$v_{\text{bias}} = \text{normalize}(\text{mean}(\text{activations}_{\text{IP}}) - \text{mean}(\text{activations}_{\text{II}}))$$

- **Loss Function:** A dual-objective loss function was used, combining the standard language modeling (LM) loss for S-CoT generation (L_{LM}) with an activation steering loss (L_{steer}) to push activations away from the bias direction.

$$L_{\text{total}} = L_{\text{LM}} + \lambda \cdot (1 - \text{cosine_similarity}(z_{\text{current}}, v_{\text{bias}}))$$

Here, z_{current} is the activation at the target layer for the current example, and λ is the steering strength (set to 1.0).

• Experiments Performed:

1. **Experiment 1 & 2:** Gemma-2-2b + S-CoT Distillation + Activation Steering (Layer 20/23). Results: 52.4% Acc, -0.035 CE (Layer 20); 49.0% Acc, 0.007 CE (Layer 23).
2. **Experiment 3:** Gemma-2-2b + S-CoT Distillation *without* Activation Steering. Results: 55.2% Acc, -0.034 CE. This isolated the effect of the steering loss.
3. **Experiment 4:** Data quality improvements (regenerating 148 incorrect traces), which yielded only marginal improvement in teacher accuracy (from 82.6% to 83.75%).
4. **Experiment 5 (Crucial Failure Point):** Gemma-3-4b-it + S-CoT Distillation + Activation Steering with the higher quality, filtered 531-example dataset.

Results and Failure Analysis (Experiment 5)

The results of Experiment 5, summarized in Table 2, represent a comprehensive failure for the simultaneous optimization approach but provided critical insights.

This failure, detailed in Table 2, stems from two primary factors. The most critical is a fundamental dual-objective conflict. While activation steering successfully reduced the Content Effect by an impressive **86%** (from 0.101 to 0.014), it came at the

Model / Experiment	Accuracy (%)	Content Effect
RoBERTa Baseline (Implicit Pattern Match)	69.86	0.101
Exp 3: Gemma-2-2b (S-CoT only, without steering)	55.20	-0.034
Exp 5: Gemma-3-4b (S-CoT + Steering, dual-objective)	49.32	0.014

Table 2: Results from Experiment 1 (Simultaneous Optimization). The dual-objective approach (Exp 5) catastrophically degrades accuracy, performing worse than both the baseline and S-CoT alone.

catastrophic cost of a **20%** drop in reasoning accuracy (from 69.86% to 49.32%). This shows the steering and reasoning losses are in direct conflict, with the former dominating the optimization. Secondly, the smaller Gemma models demonstrated a model capacity limitation. Even without steering (Exp 3), the model only achieved **55.2%** accuracy, indicating that models of this scale lack the capacity to master complex symbolic reasoning from a limited dataset of 531 examples.

4.2 Part 2: The Sequential Intervention (The SDU Approach) - The Path to Success

The comprehensive failure of Experiment 1 compelled a paradigm shift. We posited that reasoning capability and bias mitigation must be decoupled and addressed sequentially. This led to the two-stage methodology depicted in Figure 1.

4.2.1 Stage 0: Capability Acquisition - Qwen-8B S-CoT Baseline Training (Figure 1, Stage 0)

Motivation The failures with Gemma models strongly indicated a model capacity threshold. We hypothesized that a larger, more capable model (>7B parameters) would be necessary to successfully learn the complex S-CoT generation task from the distilled rationales.

Setup We selected the **Qwen-8B model** (specifically, Qwen3-8B with 4-bit NF4 quantization) as our new student model. This model possesses significantly more parameters (8 billion) compared to the Gemma models, providing increased capacity for complex pattern recognition and symbolic reasoning.

- **Training Data:** The *exact same 531 high-quality, correct S-CoT traces* that were used in the failed Gemma experiments. This ensured a direct comparison regarding the im-

pact of model capacity. The data had a fair representation for VP/VI/IP/II categories.

- **Fine-tuning Configuration:** LoRA (Low-Rank Adaptation) was employed with specific parameters:

- **Rank (r):** 8
- **Alpha (α):** 16
- **Target Modules:** ‘q_proj’, ‘k_proj’, ‘v_proj’, ‘o_proj’ (all attention projection layers, as these are critical for learning contextual representations)
- **Dropout:** 0.05

- **Training Parameters:**

- **Epochs:** 5
- **Learning Rate:** 1e-4
- **Batch Size:** 1 (with gradient accumulation over 8 steps to effectively simulate a larger batch size of 8)
- **Optimizer:** ‘paged_adamw_8bit’ with ‘bf16’ precision for memory efficiency.
- **Final Training Loss:** 0.7332.

Results (Qwen-8B S-CoT Baseline) The results, summarized in Table 3, definitively validated our capacity hypothesis.

Model	Accuracy (%)	Content Effect
RoBERTa Baseline	69.86	0.101
Gemma-3-4b (S-CoT only)	55.20	-0.034
Qwen-8B Baseline (S-CoT Distillation only)	81.50	-0.0314

Table 3: Phase 0 Results: S-CoT distillation with sufficient model capacity. The Qwen-8B model significantly outperforms both the 4B Gemma model and the RoBERTa baseline, demonstrating effective reasoning acquisition.

This constitutes a significant finding:

- **S-CoT Distillation Works with Sufficient Capacity:** The Qwen-8B model achieved **81.5% accuracy** on the test set. This dramatically outperforms the 69.86% RoBERTa baseline and the 55% accuracy of smaller Gemma models, unequivocally validating that knowledge distillation from larger teacher models (Gemini 2.5 Flash) to student models can be highly effective for complex symbolic reasoning tasks, provided the student model has adequate capacity.

- **Model Capacity is Critical:** The success of the 8B model, where 2B-4B models failed, confirms that symbolic reasoning in syllogistic tasks demands substantial model capacity. Approximately 8B+ parameters appear to be the threshold required to effectively learn and generalize intricate logical reasoning patterns from S-CoT traces.
- **Temporal Separation is Essential:** This success also implicitly validates our new sequential approach. By training the model solely on the reasoning objective (minimizing L_{LM}), we avoided the dual-objective conflict that plagued our earlier experiments.

This 81.5% accuracy model now serves as our high-capability baseline for the next stage. It exhibits a small residual bias (CE: -0.0314), meaning it has a slight tendency to favor implausible conclusions, which is the target for our SDU intervention.

4.2.2 Stage 1 & 2: SDU as a Targeted, Post-Training Intervention (Figure 1, Stage 1 & 2)

Goal To surgically correct the residual -0.0314 Content Effect of our highly accurate Qwen-8B baseline without significantly compromising its 81.5% reasoning accuracy. This is achieved through Selective Disentanglement Unlearning (SDU), adapted from Liu et al. (2024).

SDU Methodology The SDU process, applied post-training to our Qwen-8B baseline, involves three distinct sub-stages (1a, 1b, 1c and 2a, 2b, 2c in Figure 1):

Stage 1a: Shadow Model Training (Figure 1, Block 1a)

- **Purpose:** To explicitly train a model to learn the bias we intend to remove. This model will subsequently be used to identify parameters sensitive to plausibility.
- **Model:** A fresh Qwen-8B Base model (NF4 quantized) is used.
- **Training Data: 212 PLAUSIBLE examples only** (from the combined VP and IP categories of the training set). This highly focused dataset encourages the shadow model to strongly activate for plausibility.
- **Configuration:** The same LoRA setup as the baseline ($r=8$, $\alpha=16$) is used.

- **Epochs:** 3 epochs.
- **Learning Rate:** $5e-5$.
- **Final Training Loss:** 0.7820.

Stage 1b: Saliency Mask Generation (Figure 1, Blocks 1b, 1c) We identify which LoRA parameters of the *shadow model* are most responsible for encoding plausibility bias.

1. **Gradient Computation (Block 1b):** For each of the 212 plausible training examples, we compute the gradients of the shadow model's loss with respect to all LoRA parameters. These gradients indicate the sensitivity of each parameter to plausibility.
2. **Saliency Score Aggregation (Block 1b):** The absolute gradients are summed across all 212 examples to obtain a total saliency score for each of the 7,667,712 trainable LoRA parameters.
3. **Thresholding (Block 1b):** A binary **saliency mask** is created by applying a 90th percentile threshold (specific value: 2.865097) to these saliency scores. This mask identifies the top 10% of parameters deemed "most important" for encoding plausibility.
4. **Result:** This process yielded a mask covering **1,687,652 weights** out of 7,667,712 total trainable parameters, representing **22.01% coverage**.
5. **Mask Randomization (Block 1c):** To enhance robustness and prevent overfitting to specific weight locations, we randomly flip 15% of the mask bits (both $0 \rightarrow 1$ and $1 \rightarrow 0$). As suggested by Liu et al. (2024), this introduces controlled noise while preserving the overall saliency structure, aiding generalization.

Stage 2a: Bias Direction Computation (Δz) (Figure 1, Blocks 2a) To identify the precise direction in the activation space that corresponds to plausibility bias, we use contrastive examples.

1. **Data Selection:** We use **Valid-Plausible (VP)** and **Valid-Implausible (VI)** examples from the validation set. Both categories are logically valid, but they differ in conclusion plausibility, allowing us to isolate the bias.

2. **Activation Extraction:** We register forward hooks on target transformer layers (15, 18, 20, 23) in the *shadow model* (the one trained on plausible data). We then pass the VP and VI examples through this shadow model and extract the hidden state activations at these layers. The activations are averaged across all examples for each category.
3. **Bias Vector Calculation:** The bias vector Δz is computed as the difference between the mean activations of VP and VI examples. This vector precisely represents the direction in activation space that encodes plausibility bias. $\Delta z = \text{mean}(\text{activations}_{VP}) - \text{mean}(\text{activations}_{VI})$
4. **Layer-wise Bias Vector Norms:** The norms of these vectors indicate the strength of bias representation at each layer:
 - **Layer 15:** 3.50
 - **Layer 18:** 6.47
 - **Layer 20:** 8.56
 - **Layer 23:** 12.50

These norms increase with layer depth, suggesting that higher layers in the shadow model develop stronger representations of plausibility.

Stage 2b: Fine-tuning S-CoT Model (Figure 1, Arrow from Stage 0 to Correction Calc) Before applying the weight corrections, our Qwen-8B S-CoT baseline model undergoes a brief refresh.

- **Data:** 200 balanced examples (100 VALID, 100 INVALID) from the S-CoT training set.
- **Epochs:** 2 epochs.
- **Learning Rate:** $4.5e-5$ with a cosine decay schedule.
- **Purpose:** This step aims to fine-tune and refresh the model’s general reasoning representations before the surgical bias removal, ensuring the model is in an optimal state.
- **Final Loss:** 0.7890.

Stage 2c: Masked SDU Weight Corrections (Figure 1, Blocks 2b, 2c) This is the core "unlearning" step, where the Qwen-8B baseline model’s LoRA weights are selectively adjusted.

1. **Gradient-Based Correction Calculation (Block 2b):** For each LoRA parameter in the target layers (15, 18, 20, 23), a correction is computed to push the model’s activations away from the identified bias direction (Δz). The correction is formulated as: $\text{correction} = -\lambda_{SDU} \times (\Delta z \otimes \text{gradient_mask})$ where λ_{SDU} is the SDU correction strength (tested at 0.5x, 1.0x, 2.0x), and $\otimes$ denotes element-wise multiplication with the saliency mask. The ‘gradient_mask’ here refers conceptually to the saliency mask, ensuring that only bias-sensitive parameters are considered for adjustment.
2. **Selective Application (Block 2c):** Crucially, these corrections are **only applied** to the 22.01% of weights identified by the saliency masks. This surgical intervention ensures that only bias-relevant parameters are modified, safeguarding the non-biased reasoning capabilities that the model has already acquired.
3. **Multi-Strength Testing:** We experimented with three correction strengths for λ_{SDU} :
 - **Strength 0.5x (Conservative):** Corrected 78,640 weights with an average norm of 0.000068.
 - **Strength 1.0x (Standard):** Corrected 78,640 weights with an average norm of 0.000136. This yielded the optimal balance between bias reduction and accuracy preservation.
 - **Strength 2.0x (Aggressive):** Corrected 78,640 weights with an average norm of 0.000272.
4. **Target Parameters:** The corrections were applied to 32 specific LoRA parameters across the 4 identified high-bias layers (15, 18, 20, 23). This included all attention projection matrices (‘q_proj’, ‘k_proj’, ‘v_proj’, ‘o_proj’) for both ‘lora_A’ and ‘lora_B’ matrices.
5. **Final Coverage:** In total, **15% of the total trainable LoRA weights** within these target layers received bias corrections (78,640 out of 524,514 parameters).

4.2.3 Evaluation (Figure 1, Block 3)

Both the original Qwen-8B S-CoT Baseline Model (from Stage 0) and the Qwen-8B + SDU Debias

Model (from Stage 2c) are evaluated on the held-out Test Dataset. The results (Block 3) include the overall accuracy and the Content Effect, allowing for a direct comparison of the impact of the SDU intervention on both reasoning capability and bias.

5 Results of SDU Intervention and Overall Comparison

The SDU intervention was a resounding success, demonstrating the efficacy of the sequential "Train-then-Debias" paradigm. Table 4 provides a comprehensive comparison of all key models and methodologies.

Analysis of Success for Qwen-8B + SDU The Qwen-8B + SDU model represents our optimal outcome, unequivocally validating the sequential "Train-then-Debias" methodology. The SDU intervention successfully neutralized the model’s residual bias by shifting the Content Effect from -0.0314 to a near-perfect **+0.0098**, a score exceptionally close to zero that indicates the successful disentanglement of plausibility from logical judgment. This remarkable debiasing was achieved with a minimal and acceptable accuracy trade-off of only **1.4%** (dropping from 81.5% to 80.1%), highlighting SDU’s surgical precision. This result stands in stark contrast to the catastrophic **20%** accuracy loss from our initial simultaneous approach (Experiment 5), proving that a minor accuracy cost for near-zero bias is a highly successful outcome that effectively bypasses the fundamental dual-objective conflict. The positive shift from a negative CE (favoring implausible) to a slightly positive CE suggests the model is no longer systematically influenced by the content’s believability.

6 Discussion and Key Findings

Our investigation, spanning from catastrophic failure to sequential success, yields two primary findings that reshape our understanding of debiasing LLMs for complex reasoning tasks.

Finding 1: A Sequential "Train-then-Debias" Paradigm is Superior to a Flawed Dual-Objective Our investigation provides irrefutable evidence that simultaneously optimizing for reasoning (S-CoT) and debiasing (activation steering) fails catastrophically, as shown with Gemma models in Section 4.1. This dual-objective approach creates a severe conflict where the steering loss prevents the model from mastering logical structures.

Based on this insight, we validated a sequential "Train-then-Debias" paradigm that proves highly effective. First, we focused solely on capability acquisition, training a Qwen-8B model with S-CoT distillation to achieve a strong **81.5%** accuracy. Only then did we apply a surgical post-training intervention (SDU), which successfully neutralized the residual belief bias by shifting the Content Effect from **-0.0314** to a near-perfect **+0.0098**. Crucially, this was achieved with a minimal and acceptable accuracy trade-off of only **1.4%**, demonstrating that temporally separating the objectives is a superior strategy for building models that are both highly capable and robustly unbiased.

Finding 2: A Model Capacity Threshold Exists for S-CoT Distillation Our research clearly delineates a model capacity threshold for effective S-CoT distillation in symbolic reasoning tasks. While 2B and 4B parameter Gemma models failed to effectively learn the underlying logic from S-CoT traces (achieving 55% accuracy, significantly below the 69.86% RoBERTa baseline), the same S-CoT dataset led to a highly capable 81.5% accurate model when applied to an 8B parameter Qwen model. This demonstrates that S-CoT generation is a computationally demanding task that requires a substantial model capacity (approximately 8B parameters or more for this dataset). Below this threshold, models merely learn the superficial *format* of the S-CoT without grasping the intricate logical reasoning, making "shallow" pattern-matching methods more effective. This finding is crucial for guiding the selection of student models in knowledge distillation scenarios involving complex tasks.

7 Conclusion

This work demonstrates that simultaneously training LLMs to reason and unlearn bias fails due to objective conflicts. Instead, we validate a sequential "Train-then-Debias" paradigm. This sequential approach offers a robust and effective strategy for building more logical and demonstrably unbiased LLMs.

Limitations

Our study is confined to syllogistic reasoning (SemEval Task 11), and future work should validate this sequential paradigm on more diverse and complex reasoning tasks to confirm its generalizability. Furthermore, the SDU methodology is computation-

Stage	Model / Intervention	Accuracy (%)	Δ Acc from RoBERTa (%)	Content Effect (CE)	Δ CE from RoBERTa (Reduction / Shift)
RoBERTa Baseline		69.86	-	0.101	-
Failed Dual-Objective	Exp 3: Gemma-2-2b (S-CoT only)	55.20	-14.66	-0.034	-0.135 (Shift)
	Exp 5: Gemma-3-4b (S-CoT + Steering)	49.32	-20.54	0.014	-0.087 (Reduction)
Successful Sequential Paradigm	Qwen-8B Baseline (S-CoT Distillation only)	81.50	+11.64	-0.0314	-0.1324 (Shift)
	Qwen-8B + SDU (Sequential Intervention)	80.10	+10.24	+0.0098	+0.0412 (Neutralization)

Table 4: Final Results: Comprehensive comparison highlighting the catastrophic failure of the simultaneous approach versus the success of the sequential "Train-then-Debias" paradigm. SDU neutralizes bias (CE $\approx$ 0) with a minimal, acceptable 1.4% accuracy trade-off.

ally intensive, making the exploration of more efficient machine unlearning methods a valuable future direction. As we used a medium-sized (8B) model, testing this approach on larger models (e.g., 70B+) is also needed to understand the scaling laws of bias mitigation. Finally, the target layers for SDU were chosen empirically, and a more systematic investigation into optimal layer selection could yield further improvements.

References

- Ishita Dasgupta, Andrew K. Lampinen, Stephanie C. Y. Chan, Antonia Creswell, Dharshan Kumaran, James L. McClelland, and Felix Hill. 2022. [Language models show human-like content effects on reasoning](#). *Computing Research Repository*, arXiv:2207.07051.
- Jonathan St. B. T. Evans, Julie L. Barston, and Paul Pollard. 1983. On the conflict between logic and belief in syllogistic reasoning. *Memory & Cognition*, 11(3):295–306.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. In *NIPS Deep Learning and Representation Learning Workshop*.
- Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Sunkun Suda, Ranjan Kumar, and James Glass. 2023. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.
- Zhaofeng Liu, Sumit Maharjan, Fan Wu, Xiaofei Hu, and Xiang Ren. 2024. Disentangling biased knowledge from reasoning in large language models via machine unlearning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.
- Maxwell Nye, Anders Johan Andreassen, Guy Gur-Ari, Henryk Michalewski, Jacob Austin, David Bieber, David Dohan, Aitor Lewkowycz, Maarten Bosma, and David Luan. 2021. [Show your work: Scratchpads for intermediate computation with language models](#). *Computing Research Repository*, arXiv:2112.00114.
- Alex Turner, David Thiergart, Tom Rauker, Jacob Steinhardt, Atoosa Kasirzadeh, Carson Kuelbs, and Lisa Smith. 2023. [Activation addition: Steering language models without optimization](#). *Computing Research Repository*, arXiv:2308.10248.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Ren, Alexander Wang, Ziming Li, Kolter Zico, and Matt Fredrikson. 2023. [Representation engineering: A top-down approach to ai transparency](#). *Computing Research Repository*, arXiv:2310.01405.